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Numerical simulation and visualization of flow around rectangular bluff bodies

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ABSTRACT: A comprehensive study of unsteady flow around rectangular cylinders at Reynolds number of 21400 is presented in this paper. The influence of different simulation parameters, such as grid quality, turbulence model etc., are investigated in detail. Unsteady computations are performed using 2D Reynolds averaged Navier-Stokes (RANS) models and 3D large eddy simulation (LES). Time-averaged global quantities such as the mean drag coefficient, the *rms* value of lift coefficient, the Strouhal number, the recirculation length, and the surface pressure coefficient are calculated and compared with the available wind tunnel test results proposed in the literature. A series of instantaneous and time- and spanwise-average velocity, streamlines and vorticity magnitude iso-surface are also presented.

KEYWORDS: Unsteady flow; Square cylinder; Rectangular cylinder; Computational Fluid Dynamics (CFD); Large eddy simulation; Reynolds averaged Navier-Stokes;

1. INTRODUCTION

As modern long-span bridges and buildings become more flexible, the dynamic response under unsteady wind load has a greater impact on the design of these structures. Accurate prediction of wind effects on these structures is becoming increasing significant. In this respect, aerodynamic characteristics of rectangular cylinders, because of their common use in bridge and building industry, have received particular interest from both practical and academic standpoints. Bodies with a rectangular cross-section are termed as bluff bodies, in which the shear layer separated from windward corner, reattachment characteristics of the flow and vortices shedding into the wake are dependent on the cylinder's aspect ratio (width to height, *B/D*). Thus, it is important to study their aerodynamic characteristics and flow structures in details.

Okajima [1] and Tamura et al. [2] conducted detailed experimental studies to understand the aerodynamic characteristics of square cylinder and rectangular cylinders with different aspect ratios. As a consequence of the rapid growth of computation power and improvement of Computational Fluid Dynamics (CFD) technology, the numerical approach based on CFD provides a new approach for investigating aerodynamic characteristics of structures. The main purpose of the current study is to investigate to which extent a numerical simulation based on CFD can be employed to reasonably predict aerodynamic characteristics and flow structures of such a bluff body. Since the numerical approach is relatively affordable and easy to use, it is especially interesting for industrial and civil applications. Iaccarino et al. [3] examined the accuracy of RANS turbulence model in predicting complex flow with separation. The unsteady flow around square cylinder and over a wall-mounted cube were simulated and compared with experimental data. Their study demonstrated that the unsteady RANS does indeed predict periodic shedding, and leading to good concurrence with available experimental data. Shimada and Ishihara [4] numerically predicted flow features around rectangular cylinders with various aspect ratio ranging from B/D=0.6 to 8.0 by a two-layer modified $k-\varepsilon$ model. Recently, Sohankar [5] successfully reproduced the turbulent statistics of flow around rectangular cylinders with B/D=0.4~4.0 as well as the aerodynamic forces by employing unsteady and 3D large eddy simulation (LES).

In present paper, we identify a reliable CFD approach for evaluation of bluff-body flows through the simulation of flow around a square cylinder. The influence of different simulation parameters, such as grid quality, turbulence model etc., is analyzed. The numerical results are compared with the available wind tunnel test results proposed in the literature. In all the computations presented in this paper the body geometry is kept fixed and the Mach number is very low, so that compressibility effects are negligible. Furthermore, unsteady flows with zero incidence attack angle over rectangular cylinders with B/D=2.0~10.0 are numerically simulated. Also, the flow field around rectangular cylinders are analyzed in 2D and 3D space.

2. NUMERICAL METHODS

A commercial computational fluid dynamics (CFD) code, ANSYS FLUENT, was used to perform present numerical simulations. The turbulent models used in this study are the standard $k-\varepsilon$ model, RNG $k-\varepsilon$ model, realizable $k-\varepsilon$ model, standard $k-w$ model, SST $k-w$ model, RSM model, and LES model. The more detailed descriptions for all turbulent models may be found in ANSYS FLUENT Theory's Guide [6].

The ANSYS FLUENT uses the finite-volume method (FVM) to solve the governing equations for a fluid. The second order implicit scheme is used for time discretization and the second order upwind scheme is used for spatial discretization. SIMPLE (semi-implicit pressure linked equations) algorithm is employed for solving the discretized equations.

In order to obtain reliable and accurate results, it is important to choose carefully the length and width of the computational domain and boundary conditions. The computational domain and boundary conditions used in the present study are given in Figure 1. The dimensions of x_1, x_2, x_3, y_1, and y_2 adopted in the present work are summarized in Table 1. As can be seen in Table 1, for rectangular cylinders, the computational domain covers 50D~105D in streamwise direction and 30D in normal direction, which is larger than that adopted by Sohankar [5] or Sun et al. [7]. The reason for such a choice is to estimate the flow obstacle effect on the inflow and outflow boundary condition, as discussed by Murakami [8].

Considering the mesh number must be as low as possible as we could accept for efficient computation, the hybrid grids are used in the whole computational domain. The traditional finite difference method uses a structured grid, which requires a body-fitted grid transformation from physical domain to computational domain. The mesh near and aligned with the wall surfaces must be refined and stretched with the viscous boundary layer grid. The finite volume method is adopted in the present study, which makes the mesh generation flexible since it has the capacity of deal with both structured grid and unstructured grid. Figure 2 shows the grid arrangement for the rectangular cylinders. The primary characteristic of this mesh style is that the section model is nested in a rectangular area about three times larger than itself. For zone in the nesting rectangular area, the structured quadrangular grid is generated while for zones outside the nesting rectangular area, the unstructured triangular grid is applied. The height of the first layer Δy near the surface of rectangular cylinders is chosen to be 0.005D. This treatment makes it easier to generate a mesh fine enough in the vicinity of the cylinder surface while keeping the mesh in zones far away from the cylinder surface unchanged or in a proper coarser state. Details of various mesh system can be seen in Table 1. As LES simulation requires three-dimensional domain, we select the spanwise dimension to be $4D$ which is a typical spanwise length widely chosen in the other 3D simulations. The 3D grid is obtained by the structured projection of the 2D hybrid grid along the spanwise direction z. As the mean flow is two-dimensional, the spanwise grid spacing, normally taken to be uniform, is usually larger than the grid spacing in the other directions [9]. For LES simulation, we select the number of grid points N_z from 5 to 40 for discretising the spanwise length, as can be seen in Table 2.

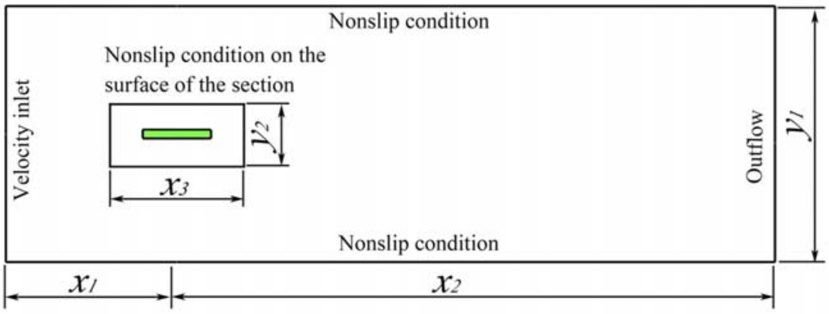


Figure 1. Computational domain and boundary conditions.

Table 1. Mesh system for present simulations.

Aspect ratio	x_1	x_2	x_3	y_1	y_2	Resolution of inner rectangular area ($N_x \times N_y$)	Grid size of computational domain ($N_x \times N_y$)
B/D=1	$10D$	$40D$	$3D$	$20D$	$3D$	50×50	50×20
B/D=2	$10D$	$40D$	$4D$	$20D$	$3D$	70×50	50×20
B/D=3	$15D$	$50D$	$5D$	$20D$	$3D$	90×50	65×20
B/D=4	$15D$	$60D$	$6D$	$20D$	$3D$	100×50	75×20
B/D=5	$20D$	$65D$	$7D$	$30D$	$3D$	120×50	85×30
B/D=6	$20D$	$65D$	$8D$	$30D$	$3D$	135×50	85×30
B/D=7	$20D$	$70D$	$9D$	$30D$	$3D$	150×50	90×30
B/D=8	$20D$	$70D$	$10D$	$30D$	$3D$	150×50	90×30
B/D=9	$25D$	$75D$	$11D$	$30D$	$3D$	150×50	100×30
B/D=10	$25D$	$80D$	$12D$	$30D$	$3D$	150×50	105×30

Table 2. 3D computational mesh parameters

Cases	Spanwise grid resolution	Grid scale
Case 1	N_z =5	$0.5D$
Case 2	N_z =10	$0.25D$
Case 3	N_z =20	$0.2D$
Case 4	N_z =30	$0.133D$
Case 5	N_z =40	$0.1D$

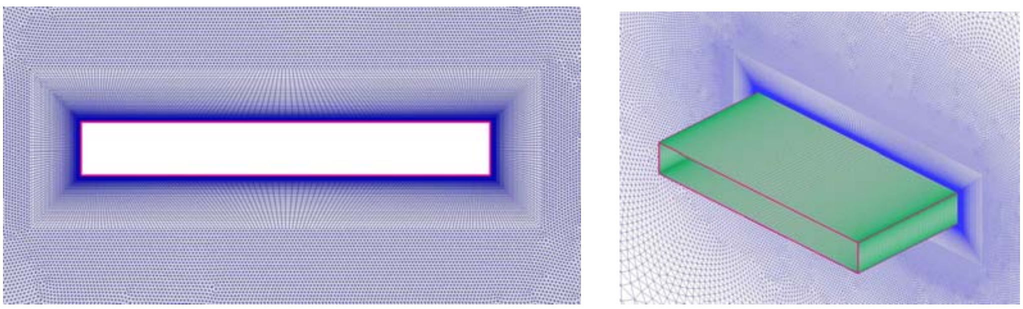


Figure 2. Grid near the surface of B/D=8 rectangular cylinder (Case 4).

The flow calculations are advanced by using a short time step (dt) to get the time dependent flow characteristics. In the present study, the time step is taken to be 0.001s. The use of smaller values of time-step did not produce any significant change in the root mean square (*rms*) and average value of the drag and lift coefficients. The time-averaged quantities for all cases have been obtained by integrating the instantaneous field over a long period of not less than 40 shedding cycles, but without including the initial transients. It is found that this period is sufficient to get a converged time-averaged flow field.

3. INFLUENCE OF MODELING PARAMETERS

3.1 *Spatial discretization*

A spanwise grid-refinement study is performed for the unsteady flow around 3D square cylinder at Reynolds number of 21400 in this section. The LES with standard Smagorinsky sub-grid model is employed. Table 3 and Figure 3 summarize typical flow parameters from the present simulation include the mean drag coefficient (C_d), the rms value of lift coefficient (*rms* C_l), the Strouhal number ($S_t=fD/U$) of the vortex shedding, and the recirculation length (l_R/D). All flow parameters are referred to the body width D. Spectral analysis was performed on the lift coefficient time history, to obtain the Strouhal number (S_t). The recirculation length (l_R/D) is the distance over which the time-averaged streamwise velocity on the rear side of the cylinder is negative. Comparisons are made with the experimental data and other numerical results in the literatures.

Table 3. Result Comparison of flow past square cylinder

Authors	Method	C_d	*rms* C_l	S_t	l_R/D
Present 2D-RANS	SKE	1.611	0.326	0.128	2.05
	RNG	1.896	1.068	0.140	0.91
	RKE	1.954	1.087	0.145	0.87
	SKW	1.956	1.502	0.131	0.80
	SST	2.130	1.435	0.126	0.91
	RSM	2.018	1.115	0.139	1.12
Present 3D-LES	Case 1	2.587	1.208	0.118	0.40
	Case 2	2.589	0.990	0.121	0.41
	Case 3	2.393	1.410	0.125	0.60
	Case 4	2.298	1.450	0.125	0.60
	Case 5	2.281	1.388	0.125	0.62
Rodi [10]	SKE	1.637	0.305	0.134	2.8
Lee [11]	SKE	1.75	–	0.138	–
	RNG	2.12	–	0.133	–
Franke and Rodi [12]	RSM	2.15	1.49	0.136	0.98
Rodi et al. [10]	LES	2.3	1.15	0.13	0.96
Lubcke et al. [13]		2.178	1.47	0.13	0.56
Bearman et al. [14]	EXP.	2.19	–	0.126	–
Lyn et al. [15]		2.10	–	0.132	0.88
Van Oudheusden et al. [16]		2.18	–	–	0.58

As can be seen in Table 3 and Figure 3, C_d is overestimated, while *rms* C_l, S_t and l_R/D are underestimated with the coarse grid. Passing to the fine grid, the accuracy of the calculated results is enhanced and a non-negligible jump is observed in C_d, *rms* C_l, S_t and l_R/D. In contrast, the results change very little passing from the fine to the finest grid. It seems possible to affirm that a reasonable convergence is reached with the fine grid, which is adopted hereafter for all the computations.

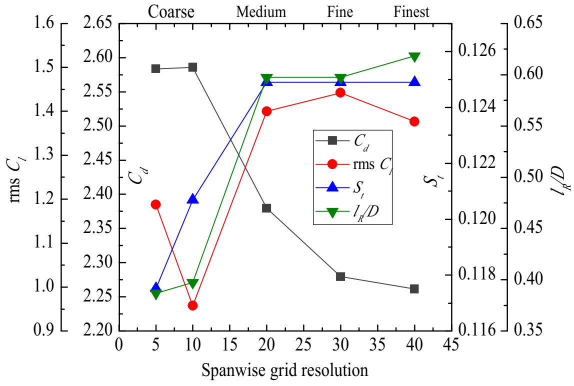


Figure 3. Grid-convergence study results

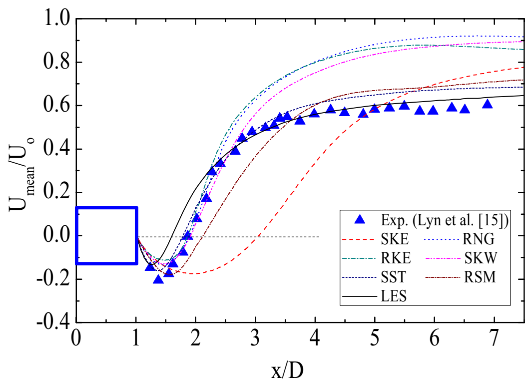


Figure 4. Averaged streamwise velocity along the centerline

3.2 *Turbulence modelling*

The influence of the turbulence model on the results is investigated in this section, and the simulated flow parameters by various turbulence models are also summarized in Table 3. From Table 3, we find that C_d and *rms* C_l predicted by all 2D RANS and 3D LES turbulence models agree well with the experiments and also with the other numerical results. Comparatively speaking, the standard $k-\varepsilon$ model underestimate C_d and *rms* C_l considerably. Most RANS calculations and LES with fine grid yield the correct value of S_t=0.125-0.140, and it appears that St is not very sensitive to the choices of turbulence model. The recirculation length varies most widely with different choices of turbulence model. The experiment and most numerical simulations yield the value of l_R/D≈0.56-0.96. The RSM model overestimates the value of l_R/D slightly while the standard $k-\varepsilon$ model gives the incorrect quantity of l_R/D. These similar results was also found in the calculations of Rodi [10] and Lee [11] in simulating the flow around the square cylinder. It indicates that the standard $k-\varepsilon$ model is not capable of simulating complex unsteady turbulent flow around a bluff body.

The distributions of the time-averaged streamwise velocity U_{mean} along the centerline are displayed in Figure 4. There the results reflect the recirculation length l_R/D behavior discussed already. The standard $k-\varepsilon$ model overpredicts the length of the recirculation zone considerably; introducing the RNG and Realizable modifications improves the calculations. The approach to the free-stream velocity is however faster than the measurement by Lyn et al. [15]. Turning to the $k-\omega$ calculations, it can be seen from Figure 4 that the standard $k-\omega$ model show an unrealistically distributions of U_{mean} along the centerline while the SST model show better behavior. In the near-wake region, the SST calculations show very good agreement with experiments. The SST results are even superior to those obtained by much more complex RSM model which overestimates slightly the recirculation region. Although the LES model underestimates the recirculation region, it approach the exact free-stream velocity compared with Lyn's measurement.

The distribution of the *rms* value of velocity along the centerline, shown in Figure 5, is determined to be too low if the standard $k-\varepsilon$ model is used. This is due to the overproduction of turbulent kinetic energy in front of the cylinder, resulting in a too strong damping of the periodic shedding motion in the wake of cylinder. It should mentioned that there are considerable differences to the calculations of Rodi [10], who with the same model, obtain a much higher lever of fluctuations. This indicates that the results are sensitive to the details of the simulation parameters. The peak values of fluctuations due to the revised $k-\varepsilon$ models, RNG and Realizable, show slight closing to the experiments. The standard $k-\omega$ model underpredicts the *rms* streamwise velocity, while the SST and RSM models overpredicts this quantity. The LES computation gives approximately the correct level and distribution of *rms* streamwise velocity. Turning to the *rms* lateral velocity, the standard $k-\omega$ and RSM models also give an unrealistic level and distribution of this quantity, while the SST and LES models improve the calculations. Due to the underestimation of recirculation region, the LES model yields the pick value of *rms* lateral velocity a little to the left.

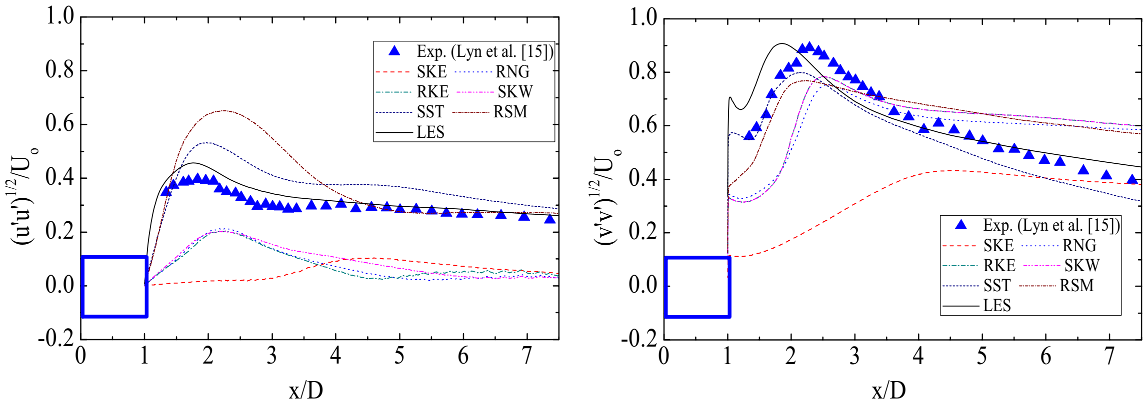


Figure 5. *rms* streamwise velocity and *rms* lateral velocity along the centerline

4. EFFECT OF ASPECT RATIO

The turbulent flow around rectangular cylinders with different aspect ratios ranging from *B/D*=2.0 to *B/D*=10.0 are simulated using SST $k-w$ and LES model in this section. A Reynolds number of 21400 is again performed. Mean drag coefficients, Strouhal number, and *rms* value of lift coefficients of present work are compared in Figure 7 with the wind tunnel experimental results and results of other 3D LES simulations.

In general, the present SST and LES results are almost consistent with both the experimental results and other numerical analysis. The mean drag coefficients and the fluctuating lift coefficients decrease monotonically with increasing aspect ratio. According to the experiments conducted by Okajima (1983), the Strouhal number has a minimum value around *B/D*=2.0, and a maximum around *B/D*=3.0. This fact is well reproduced in the present simulation. Figure 8 shows the streamlines of sections with typical aspect ratios of 2.0, 3.0, 5.0 and 8.0 predicted by SST model. As can be seen in Figure 8, the recirculation vortices are generated near the upper and lower face of the *B/D*=2.0 rectangular cylinder, and the flow separated at the leading edge is not reattached to the wall. In addition, the location of the generated of the vortex behind the cross-section is far away from the surface, and result in the relatively small of Strouhal number. For *B/D*=3.0 and *B/D*=5.0 rectangular cylinder, the flow separation at the leading edge is developed and the separated flow cannot detach themselves fully from the surface but reattach above or below the section during a period of vortex shedding. Also, as can be observed in Figure 8(b) and Figure 8(c), the intermittently reattachment, which results in a narrowing of the wake, accompanied by the discontinuity in Strouhal number between *B/D*=2.0 and *B/D*=3.0. Another discontinuity in Strouhal number can be recognized between *B/D*=6.0 and *B/D*=8.0. From the streamlines plot in Figure 8(d), there is a steady reattachment just behind the leading edge of *B/D*=8.0 rectangular cylinder, and flow finally separates at the trailing edge. The transition of the separated flow from a periodical reattachment type to a steady reattachment type, which leads to a increase in Strouhal number, is predicted well in present calculations. As the aerodynamic behavior of rectangular cylinders with different aspect ratios is characterized by the absence or presence of reattachment of the shear layer, the rectangular cylinders considered in present simulation is divided into three categories: separated type (B/D=1.0 and B/D=2.0), intermittently reattached type (3.0≤B/D≤6.0), and fully reattached type (7.0≤B/D≤10.0).

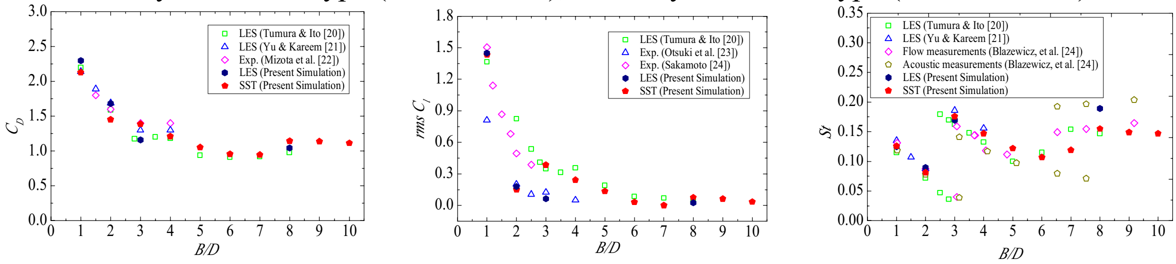
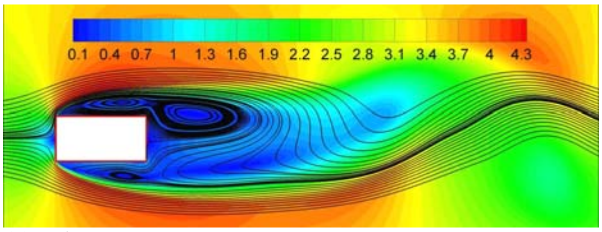
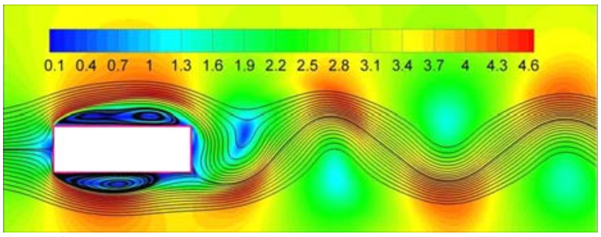


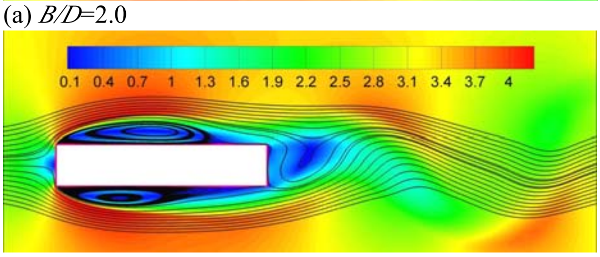
Figure 7. Variation of mean drag coefficients (C_d), Strouhal number (S_t) and *rms* lift coefficients (*rms* C_l) with aspect ratio (B/D), Re=21400



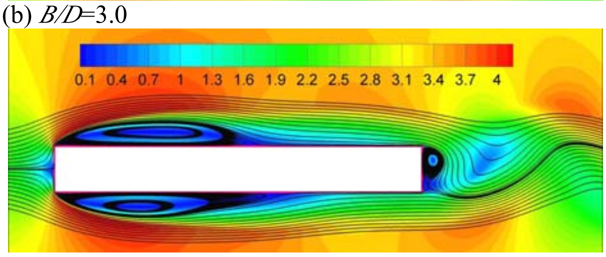
(a) B/D=2.0



(b) B/D=3.0



(c) B/D=5.0

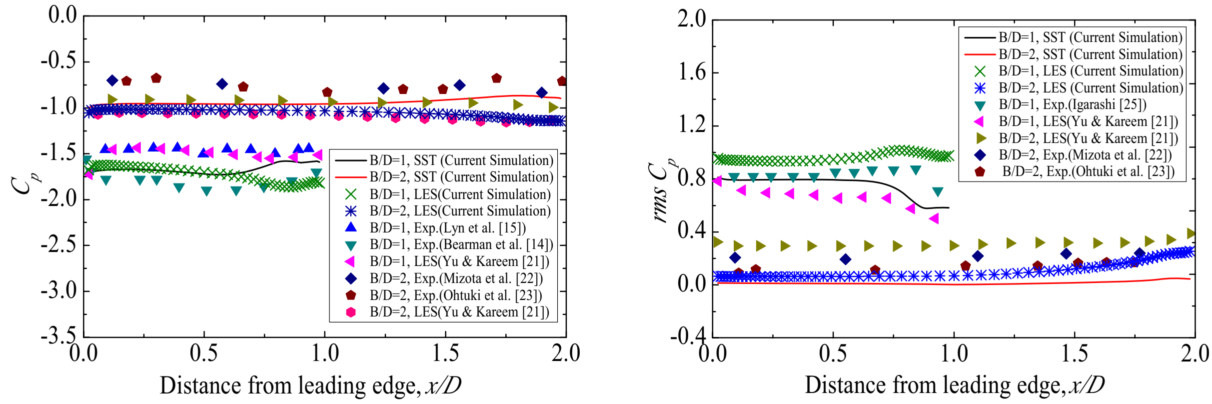


(d) B/D=8.0

Figure 8. Instantaneous velocity contours with streamlines

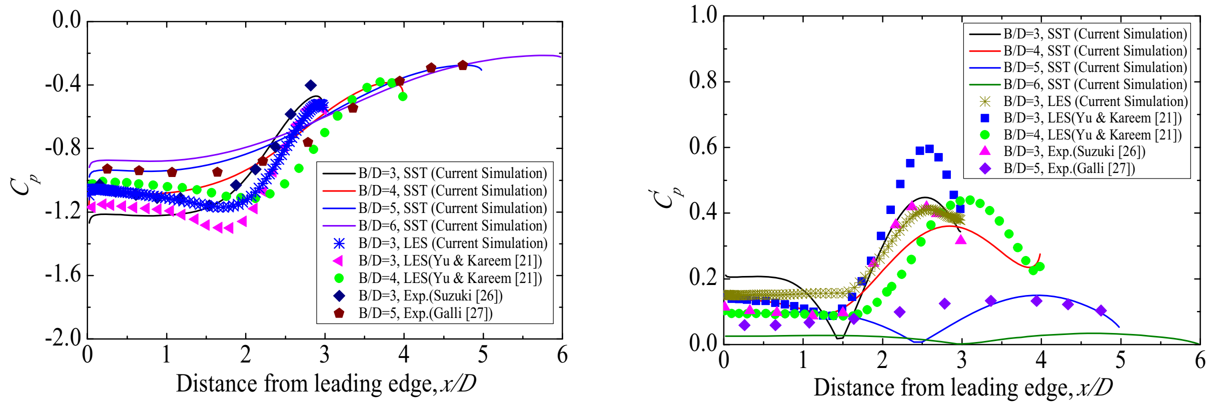
In Figure 9(a), 10(a) and 11(a), the mean pressure distribution on the side of rectangular cylinders with different aspect ratios is presented. The x-axis in the figure is normalized with the cylinder width. For separated-type rectangular cylinders, the mean pressure remains almost constant along the cylinder surface. The pressure on the side face of B/D=2.0 rectangular cylinder are higher (less negative) than those of a square cylinder, due to the influence of the afterbody on the separated flow. With an increase in the streamwise body width, i.e., aspect ratio, there is a concomitant pressure recovery on the side face due to the reattachment of the flow. Furthermore, the present LES and SST simulations are in good agreement with the experimental data and other LES results.

In Figure 8(b), 10(b) and 11(b), the *rms* pressure fluctuations on the side of rectangular cylinders with different aspect ratios are compared, revealing that for square cylinder and B/D=2.0 rectangular cylinder, a local minimum on the side surface does not exist, but for 3.0≤B/D≤10.0 rectangular cylinders minima are observed. This is also due to the reattachment of the flow. In general, the present simulated *rms* pressure fluctuations are in good agreement with the experimental data and other LES results. However, the present SST simulations slightly underestimate the *rms* pressure fluctuations compared to LES data and experimental results. It should be noted that the total pressure fluctuation includes periodic fluctuation component and stochastic fluctuation component of the pressure, and the stochastic fluctuation component is not modelled explicitly in any current RANS models. Thus, its value is always underestimated compared to the exact value of its fluctuation.



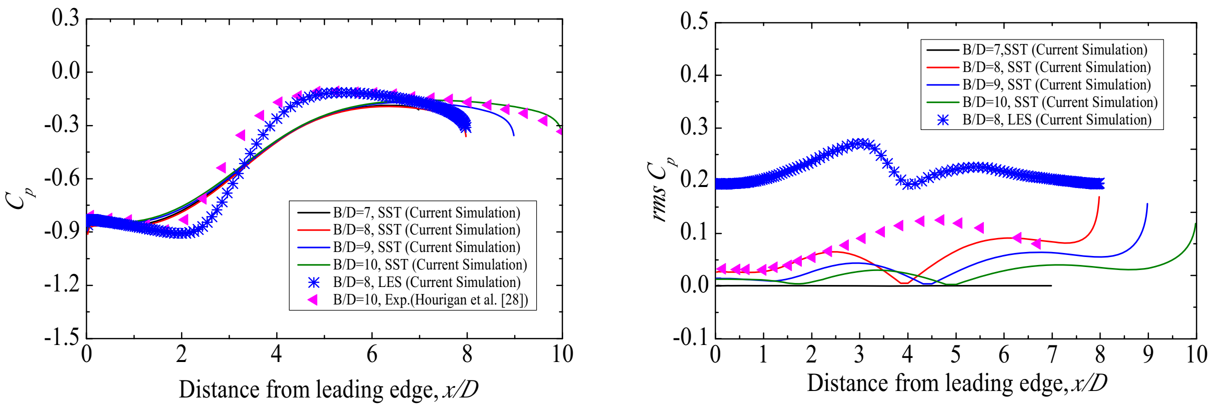
(a) Mean pressure distribution (b) *rms* pressure fluctuations distribution

Figure 9. Comparison of pressure distribution on the side of separated type rectangular cylinders



(a) Mean pressure distribution (b) *rms* pressure fluctuations distribution

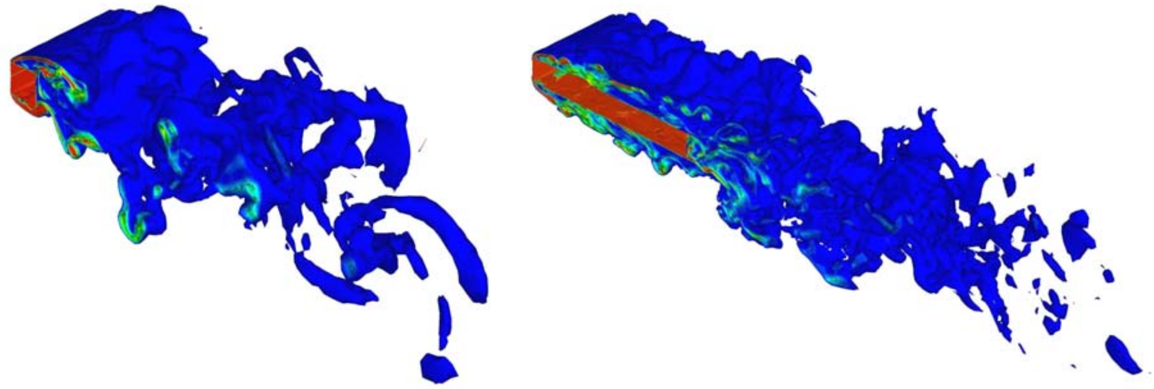
Figure 10. Comparison of pressure distribution on the side of intermittently reattached type rectangular cylinders



(a) Mean pressure distribution (b) *rms* pressure fluctuations distribution

Figure 11. Comparison of pressure distribution on the side of fully reattached type rectangular cylinders

The 3D flow field around various rectangular cylinders are obtained by LES simulations and the instantaneous vorticity magnitude iso-surface of square cylinder and B/D=8.0 rectangular cylinder are shown in Figure 12. From a qualitative point of view, 3D flow structures clearly appear even though the flow field 2D main features. Generally speaking, the further the location from the separation point at the leading edge, the more significant the 3D features of the flow structures. The wake structure of square cylinder and *B/D*=8.0 rectangular cylinder is irregular and 3D, as clearly seen in Figure 12.



(a) Square cylinder (b) *B/D*=8.0 rectangular cylinder
Figure 12. 3D instantaneous vorticity magnitude

4. CONCLUSIONS

Numerical simulation results of unsteady flows with zero incidence attack angle over rectangular cylinders at Reynolds number of 21400 has been presented in this paper. Despite the simple geometry of the obstacle, the flow develop in its vicinity is very complex with multiple, unsteady flow, various kinds of vortices and strong curvature. The effectiveness of the turbulence models and numerical treatment for simulating the bluff-body flows are investigated in detail. Some partial conclusions can be made:

(1) The spanwise grid dependent results give an idea of the degree of grid refinement necessary to correctly simulate the 3D flows around bluff bodies.
(2) Seven turbulence models have been tested: the 3D LES model can provide satisfactory prediction for the unsteady flow field features of a square cylinder. 2D RANS approach is able to predict the main features of this complex flow reasonably well, while at the same time not being very CPU demanding. The SST $k-\omega$ model is found to be the best choice among RANS models and has accuracy enough to be suitable for practical problems.
(3) Based on the simulated results of the flow around rectangular cylinders with aspect ratio ranging from *B/D*=2.0 to 10.0, it is found that the mean drag coefficients and the fluctuating lift coefficients decrease with increasing aspect ratio. Between *B/D*=2.0 and 3.0, a discontinuity in Strouhal number occurs duo to intermittently reattachment of separated flow. When the aspect ratio *B/D* reaches to about 7.0, a steady reattachment behind the leading edge occurs, which leads to a increase in Strouhal number. The pressure coefficients distribution of rectangular cylinders, simulated by present 2D SST model and 3D LES model, are found to be almost consistent with both the experimental results.

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